

Fardin Ahmed

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PROFESSIONAL SUMMARY

Computer Science student at Georgia Tech with software engineering internship experience at Cisco and SAIC, building production AI tooling, C++ backend systems, and full-stack applications. Proven ability to deliver measurable results with a foundation spanning systems programming, machine learning, and security.

EDUCATION

Georgia Institute of Technology
B.S. in Computer Science.

Atlanta, GA
May 2027

SKILLS

Languages: Python, Java, C/C++, JavaScript, Go, SQL, MATLAB, R, HTML/CSS

Frameworks: React, Node.js/Express, Django, Flask

Libraries: PyTorch, NumPy, Google Test, Matplotlib, SFML, Pygame, Libgdx

Developer Tools: Git, AWS, Jira, Shell Scripting, VirtualBox, Makefile, Eclipse, Cursor

Operating Systems: Linux (Debian, Ubuntu, Red Hat), MacOS, Windows

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Cisco Systems, Inc.

San Jose, CA

- Designing and shipping an end-to-end MCP-based CLI agent in stdlib-only Python that exposes live switch diagnostics as structured tools in air-gapped environments, running a quantized 2B-parameter Gemma 4 model via llama.cpp; adopted as an internal network operations tool to reduce manual CLI troubleshooting.
- Engineered a ReAct reasoning loop with self-evaluation, checkpoint backtracking, and BM25 retrieval over 851 SONiC and Cisco-specific command references; distilled tool-use trajectories from a larger teacher model for LoRA fine-tuning, increasing accurate tool-calling and argument generation from 60% to 92%.
- Built 112 automated test modules and a read-only command validator blocking write and mutating operations; optimized the containerized inference stack with KV-cache quantization, CPU pinning, and memory tuning, reducing memory usage by 42% from 4.5 GiB to 2.6 GiB.

Undergraduate Researcher, VIP Program

January 2026 – Present

Georgia Institute of Technology

Atlanta, GA

- Modified and pretrained TinyTimeMixer (TTM) by increasing hidden dimensions from 64 to 128, reducing FRED-MD forecasting MSE by 83% from 1.826 to 0.315.
- Benchmarked Chronos Bolt Tiny, PatchTST, DeepAR, and TTM on 126 FRED-MD macroeconomic series using RMSE, MAE, and MASE metrics via AutoGluon.

Software Engineering Intern

May 2025 – August 2025

Science Applications International Corporation (SAIC)

Colorado Springs, CO

- Contributed to development and testing of critical defense software systems, including radar system components, in a high-stakes Agile environment while holding an Interim Secret clearance.
- Modernized object-oriented C++ backend modules by transitioning legacy communication protocols to TCP/IP socket-based systems; implemented 50+ Google Test unit tests, raising test coverage from 15% to 35%.

PROJECTS

Safelight <https://safelighted.web.app/>

April 2026 – Present

React, Node/Express, PostgreSQL

- Designed and implemented end-to-end encryption for a full-stack messaging app, building a zero-dependency Web Crypto module securing every message with per-message X25519 ECDH (256-bit keys), HKDF-SHA256, and AES-256-GCM in a multi-recipient sealed-envelope scheme.
- Engineered a client-side password-derived key vault using PBKDF2 (310,000 iterations) and AES-256-GCM; added SHA-512 (5,200-round) safety-number verification and a 10-case automated suite covering tamper, wrong-key, and replay rejection.

OmniDetect <https://github.com/FinEredium1/OmniDetect>

January 2026 – May 2026

Python, scikit-learn, PyTorch, librosa

- Built the text-detection component of a 5-person team project flagging AI-generated content, engineering a TF-IDF pipeline (single and two-word phrases, 70,000-word vocabulary) and training a LinearSVC classifier that reached 96.93% accuracy at a 0.8% false-positive rate.
- Validated generalization with leakage-safe 70/15/15 splits and 4 controlled baseline experiments, sustaining 95.6% accuracy on a separate 464K-sample unseen dataset.